



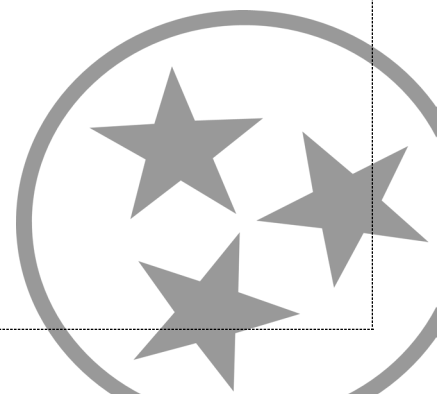
Department of
Health



Controlled Substance Monitoring Database

2025 Report to the 114th Tennessee General Assembly

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Tennessee Department of Health Key Points Related to the Drug Epidemic

The Tennessee (TN) Controlled Substance Monitoring Database (CSMD) is a Prescription Drug Monitoring Program (PDMP) designed to provide healthcare practitioners with a comprehensive view of a patient’s history of controlled substance prescriptions. The CSMD contains prescription information from all dispensers of controlled substances in Tennessee, including Veterans Health Administration (VHA) pharmacies in Tennessee. The CSMD collects and maintains dispensing data regarding all controlled substances in Schedules II, III, IV, and V controlled substances.

The purpose of the CSMD is to increase the quality of patient care by equipping healthcare practitioners with accurate, timely information that the practitioners can use to determine when patients acquiring controlled substances may require counseling or intervention for substance abuse, by collecting and maintaining data regarding all controlled substances in Schedules II, III, and IV dispensed in this state, and Schedule V controlled substances identified by the controlled substance database committee as demonstrating a potential for abuse. Further, the database is to be used to assist in research, statistical analysis, criminal investigations, enforcement of standards of health professional practice, and state or federal laws involving controlled substances.

In accordance with the Controlled Substance Monitoring Act of 2002, the CSMD was established. Data collection began for all dispensers on December 1, 2006. The Prescription Safety Acts (PSA) of 2012 and 2016 enhanced the monitoring capabilities of the database especially with mandatory registration and use starting in 2013. The CSMD became timelier and more meaningful in 2016 when data for human patients had to be submitted at least once every business day for all the controlled substances dispensed, but no later than the close of business on the following business day. Additional changes to the laws affecting the database were made by Tennessee Together legislation, which increased the frequency for mandatory checks of the CSMD for prescribing and dispensing of opioids and benzodiazepines from every twelve (12) months to every six (6) months. Prescribers and pharmacists can access prescription information from greater than half of the states in the US, in addition to the US Department of Defense.

No rulemaking amendments were necessary and/or required for the CSMD in 2024. However, in 2023, the General Assembly extended the TN Together opioid initiative in Public Chapter 188 by deleting Section 9 of 2018 Public Chapter 1039 which originally provided for sections of the Act to expire on July 1, 2023. In addition to extending the Act, the General Assembly also added a definition for “recent cancer treatment” and excepts patients from the Act who have undergone recent cancer treatment or who are currently undergoing treatment. The General Assembly also removed the need for an opioid prescriber to obtain updated, periodic informed consent from a patient when writing a subsequent prescription for the same opioid and for the same episode of treatment. Lastly, the General Assembly also amended the reporting requirements under this law such that the Commissioner of Health shall periodically provide an update to the General Assembly via letter regarding the impact and effect of this Act.

This 2024 CSMD report is designed to provide the General Assembly with an update on activities and outcomes related to the substance abuse crisis as it pertains to the CSMD and the Tennessee Department of Health (TDH). The CSMD Committee reports annually on the outcome of the program with respect to its effect on distribution and abuse of controlled substances, along with recommendations for improving control, prevention, and minimizing diversion of controlled substances.

PDMPs, like Tennessee's CSMD, are the cornerstone to state-level interventions to improve prescribing of controlled substances, to inform clinical practice, and to protect at-risk patients. Provision of accurate and timely dispensing information is essential to critical clinical decision making which can provide safe and effective treatment of pain.

Key Analytical Findings:

- Morphine Milligram Equivalents (MME) prescribed and dispensed to patients in TN has decreased by 54.2%. (2017-2024)
- Number of potential doctor shoppers has decreased by 79%. (2017-2024)
- Number of opioid prescriptions for pain has decreased by 43.1%. (2017-2024)
- Number of Gateway integrations (in production) has increased by 44.3%. (2023-2024)

Key Highlights for CSMD in 2024:

- The number of controlled substance prescriptions reported to the CSMD has remained relatively unchanged.
- Response time for searches in the CSMD was less than one second if the request did not include data from another state.
- Gateway integration with Electronic Health Record (EHR) and Pharmacy Management System workflows has increased significantly across the state enabling users to provide more direct-to-patient care.
- The total number of patient queries increased by 31% from 2023.
- The number of patient reports increased by 11.9% from 2023.

Considerations for 2025:

- Compared to 2022, Tennessee saw a decrease (5.5%) for fatal overdoses in 2023.
- While increases in overdoses continue to be primarily related to illicit substances such as fentanyl and methamphetamine, inappropriate use of prescription drugs still poses a major risk to Tennesseans.
- Opioids typically obtained through a prescription for pain relief were involved in 422 overdose deaths in 2023, a 19% decrease from 2022.
- Among all opioid-related overdose deaths in 2023, 13% also involved benzodiazepines, highlighting the critical importance of avoiding concomitant use of opioids and benzodiazepines.
- In 2023, 41% who died of drug overdose involving opioids and benzodiazepines had a controlled substance dispensed within 60 days of death. Among people who died of an overdose involving pain relievers, 30% had an opioid prescription. Among those who died of an overdose involving benzodiazepines, 28% had a benzodiazepine prescription within 60 days of death.
- The combination of prescription and illicit substances cause of death reinforces the need for a three-pronged approach of prevention, treatment, and enforcement.

Trends in Drug Overdose Deaths in Tennessee and the Role of the CSMD

TDH uses methodology established by the Centers for Disease Control and Prevention (CDC) to understand and describe drug overdose deaths in our state (CDC, 2016)¹. Data from Vital Statistics indicates from 2022 to 2023, drug overdose deaths in Tennessee declined by 5.5%, decreasing from 3,826 to 3,616.² Eighty-one percent (81%) of drug overdose deaths in 2023 involved opioids. Opioid overdose deaths are primarily driven by illicit fentanyl. However, the number of overdose deaths in which fentanyl was involved decreased 2.8% between 2022 and 2023, from 2,797 to 2,720. Fentanyl still accounts for 75% of all drug overdose deaths in 2023. Deaths involving heroin have consistently decreased from 2019 to 2023, accounting for 0.9% of all overdose deaths in 2023. Deaths involving opioids identified as pain relievers (i.e., those typically obtained through a prescription) also have a declining trend. In 2020, pain relievers accounted for 19.6% of overdose deaths. In 2023, pain relievers accounted 11.7% of overdose deaths, nearly an 8% decline.

Deaths involving any stimulants have consistently increased over the past five years: a 143% increase from 2019 to 2023. The proportion of deaths involving stimulants other than cocaine, a category that includes primarily deaths involving methamphetamine, have increased substantially over this period (42% of all overdose deaths in 2023). Deaths involving both opioids and stimulants have also increased over the past five years. In 2023, 79% of stimulant-involved deaths also involved an opioid. Of these deaths (n=1,748), the opioid most frequently involved was fentanyl (96%). Deaths associated with benzodiazepines increased slightly from 2022 (12% of all overdose deaths in 2023). Deaths that included a combination of benzodiazepines and opioids increased by 3% from 2022 to 2023. In 2023, 13% of opioid associated deaths also included a benzodiazepine.

Concomitant improvements in a number of measures of good medical practice, including reductions in the number of opioids prescribed and dispensed, fewer doctor shoppers, and increased utilization of the CSMD suggest that increased awareness among the medical community and statewide interventions may have lessened the impact of prescription drugs on overdose mortality. Among individuals who died of drug overdose in 2023, 22% had a controlled substance dispensed within 60 days of death, a decrease from 24% the year before. This downward trend has been consistent, year over year since 2013, and suggests that other factors are playing a significant role in the increasing opioid overdose deaths, such as illicit fentanyl, stimulants, and diverted prescription opioids.

TDH is committed to improving the way CSMD data are used to help curb the epidemic of overdose in Tennessee. The Office of Informatics and Analytics (OIA) maintains the Integrated Data System (IDS) which combines data from the CSMD with other patient health data to identify key markers for increased risk of overdose. Epidemiologists at TDH have conducted a number of studies and are developing several tools using these linked data to better understand patient trajectories as they move from prescription drugs into the illicit market, and to better understand what puts Tennesseans at higher risk for overdose and death. With these data, policy and prevention, and intervention programs, treatment can be targeted more specifically to intervene early, when recovery is easier and more likely to be successful.

In addition, OIA has developed a data driven method of identifying prescribers who may be engaging in high risk prescribing or who have high risk patient populations. The first high risk prescriber lists were created in 2019, and the method continues to undergo refinement to better identify risky prescribing. Work also continues to identify patients

¹ Rudd RA, Seth P, David F, Scholl L. Increases in Drug and Opioid-Involved Overdose Deaths — United States, 2010–2015. *MMWR Morb Mortal Wkly Rep* 2016;65:1445–1452. DOI: <http://dx.doi.org/10.15585/mmwr.mm65051e1>

² 2022 Tennessee Drug Overdose Death Report. https://www.tn.gov/content/dam/tn/health/documents/pdo/death-report/2022_Tennessee_Drug_Overdose_Deaths.pdf

at high risk of overdose, and the department has partnered with researchers at Vanderbilt University Medical Center (VUMC) to apply advanced machine learning techniques to better predict overdose risk among Tennesseans. The initial phase of this work has concluded. In the coming year, TDH also plans to launch enhanced prescriber reports to registered CSMD users that leverage the power of OIA's analytics and data linkage efforts to provide prescribing and overdose information back to Tennessee's prescribers.

TDH is working closely with a number of other departments, including the Tennessee Department of Mental Health and Substance Abuse Services (TDMHSAS) and the Tennessee Bureau of Investigation (TBI), to respond to the epidemic. Through these partnerships, TDH is providing county-level data to stakeholders in communities across Tennessee. Overdose and controlled substance prescribing data has been invaluable for planning and resource allocation for TDH and TDMHSAS prevention and response projects.

OIA produces several public annual reports that present data on overdoses and prescribing throughout Tennessee that are greatly enhanced by the addition of CSMD data.

- The annual Tennessee Drug Overdose Death Report includes information on fatal overdose and decedent prescription histories using CSMD data and is released when data are finalized usually between September and December.
- The annual Drug Overdose Hospital Discharge Report includes information on nonfatal overdoses for patients who were seen at a TN hospital and is released on March 1.
- These reports, including associated slides and infographics, can be found at: <https://www.tn.gov/health/health-program-areas/pdo/pdo/facts-figures.html>. In late 2020, OIA released a special report on buprenorphine to provide an in depth look at prescription and patient trends as well as buprenorphine involvement in fatal overdoses and buprenorphine prescribing before and after nonfatal overdoses.

OIA also produces an interactive online dashboard that includes overdose and prescription information and can be found at this address: <https://www.tn.gov/health/health-program-areas/pdo/pdo/data-dashboard.html>. The drug overdose data dashboard underwent a significant overhaul to enhance its usefulness and visual appeal and relaunched in late summer 2020. The new version of the dashboard was designed to accommodate more frequent updates to the data and content.

Weekly Hospital and Emergency Department Data

In 2023, for every drug overdose death, there are approximately seven (7) nonfatal overdoses as identified in discharge data from emergency departments and hospitals. The proportion of these hospital visits due to opioids has steadily increased, primarily driven by a substantial increase in fentanyl related nonfatal overdoses in recent years.³ OIA estimates at least 20% of overdose decedents in 2023 had a nonfatal overdose in the year before their death (not published).

These overdoses are treated in emergency departments and hospitals, but information about overdoses is not currently available to clinicians outside the hospital or through the CSMD. In 2016, Public Chapter 959 provided the Commissioner with the opportunity to require healthcare facilities to provide TDH with near real-time data on nonfatal drug overdoses. Such a data collection system was implemented in 2017, with a pilot project involving 11 hospitals. The Drug Overdose Reporting system (DOR) is now in its active stage, with 117 hospitals reporting to

TDH from across the state. Opioid drug overdose was included on the Tennessee Reportable Disease List in 2019.³ From the pilot stage through 2019, hospitals have only been required to report opioid overdoses. The Tennessee Reportable Disease List was expanded in 2020, and hospitals are now asked to report overdoses involving a number of other substances of concern, specifically stimulants, benzodiazepines, and muscle relaxants. In 2021, new opioid codes were expanded to capture fentanyl and tramadol. In 2022, The Tennessee Reportable Disease List was expanded again, and hospitals are now asked to report overdoses involving cannabis and hallucinogens to capture overdose more accurately across the state. This expanded range of reporting provided better insights into the changing nature of Tennessee's overdose epidemic and potentially reveal timely trends in overdoses that equip the state to respond to new and emerging threats.

Preliminary estimates show that overdoses seen in a hospital are reported on average, just over a week after the patient is discharged, making DOR one of the fastest sources of overdose information available to the department. This speed makes DOR data one of the department's most important tools for planning and resource allocation for overdose response. Overdoses reported to DOR are used in several regular data briefs that are shared with regional epidemiologists, including information on those overdoses associated with active CSMD prescriptions. As of Fall 2019, record-level DOR data have been made available to regional and metro health departments to guide local response efforts. In Fall 2020, TDH began to make monthly reports on statewide DOR data available to the public.⁴ Additionally, TDH convenes a monthly multi-agency workgroup made up of several divisions across TDH, the TDMHSAS, the TBI, and others to discuss current temporal and geographic trends in the overdose data. These meetings play a vital role in state agency situational awareness and offer a valuable opportunity for state overdose stakeholders to share challenges, successes, and support.

TDH does not currently have a reliable estimate of the number of overdoses which are managed in the field where the patient refuses transport to the hospital. TDH expects that this represents an even larger number of nonfatal overdoses that are not currently being systematically tracked. In order to overcome this gap, the OIA is actively working with the Office of Emergency Medical Services to obtain statewide emergency medical service data on overdoses seen in the field. Once these data are available, TDH anticipates creating a number of reports and public health surveillance products that will better inform stakeholders about the prevalence and trends in nonfatal overdose throughout the state.

TDH continues to explore ways to provide patient overdose information from these sources back to providers so that they might have additional patient history data to make better informed decisions about opioid prescribing. In 2023, TDH plans to launch enhanced prescriber reports to registered users of the CSMD. These reports are expected to include information on patients of the prescribers who have overdoses that have been reported to TDH.

Neonatal Abstinence Syndrome (NAS) Surveillance Update

In 2022, 1.0% (n=797) of live births were diagnosed with NAS, an increase from 0.9% of live births (n=758) in 2021. Similar to previous years, the highest percentage of cases (62%) were exposed to Medications for Opioid Use Disorder treatment (MOUD). MOUD plus therapy is the standard of care for pregnant women with Opioid Use

³ Tennessee Reportable Disease List. <https://www.tn.gov/health/cedep/reportable-diseases.html>

⁴ Drug Overdose Facts & Figures. <https://www.tn.gov/health/health-program-areas/pdo/pdo/facts-figures.html>

Disorder (OUD). Regional variations of reported NAS births align with the known geographic distribution of buprenorphine prescriptions for MAT.⁵

Number of Registrants in the CSMD

The PSA of 2012 facilitated a substantial increase in CSMD utilization, and the PSA of 2016 and other legislation further expanded the requirement for a mandatory CSMD check by healthcare practitioners. Year after year, the CSMD continues to have moderate increases in the number of registrants. By the end of 2024, the number of registrants had grown to 68,154, an increase of 2.9% from 2023.

Number of Registrants in the CSMD, 2017-2024

Number of Active Registrants in the CSMD, 2017 - 2024*		
Year	Registrants	Change (%)
2017	47,294	-
2018	50,991	7.8
2019	54,642	7.2
2020	58,379	6.8
2021	61,390	5.2
2022	64,717	5.4
2023	66,226	2.3
2024	68,154	2.9
*VA registrants were included.		

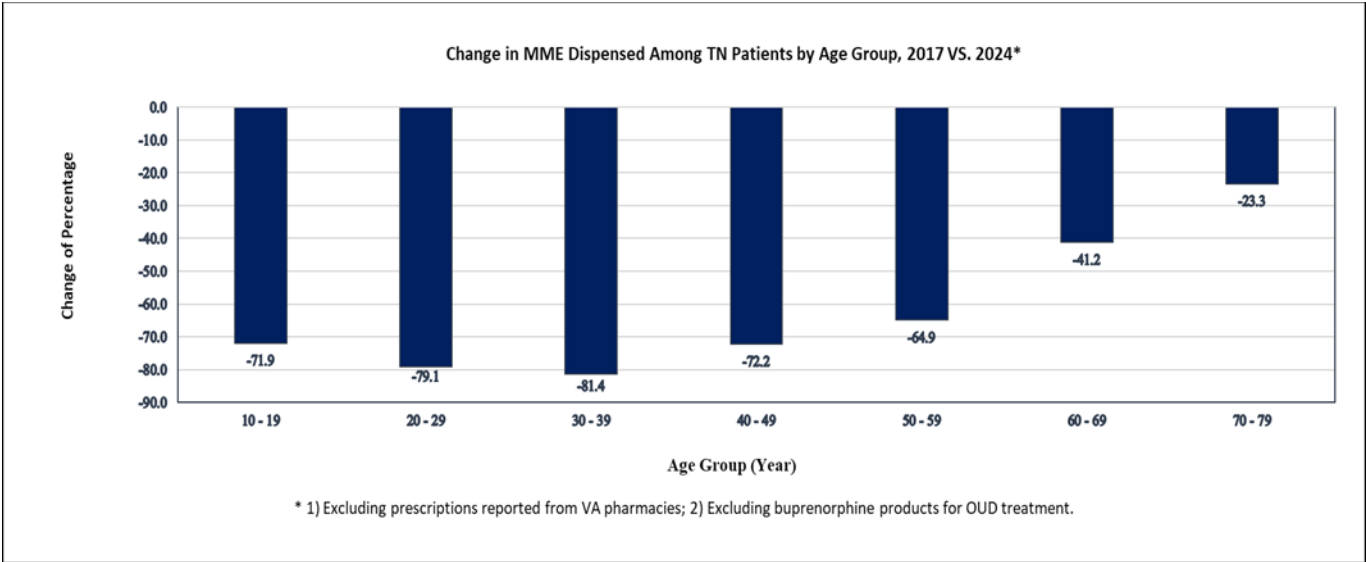
Law enforcement requests to the CSMD continue to be a critical use of the CSMD as TDH works together to address questionable controlled substance use in TN. During 2024, there were 5,149 law enforcement related requests to the CSMD.

MME Improvements and Concerns by Age Group

The OIA provides an analysis of the MME for TN patients by age groups. Encouragingly, there was a decline in MME dispensed for patients in the various age groups below comparing 2017 to 2024. Improvements in these age groups are an indicator that TDH’s efforts are successfully preventing individuals from being overexposed to opioids by the healthcare system.

⁵ TN Buprenorphine Report.
https://www.tn.gov/content/dam/tn/health/documents/pdo/2020%20Buprenorphine%20Report_11.30.pdf

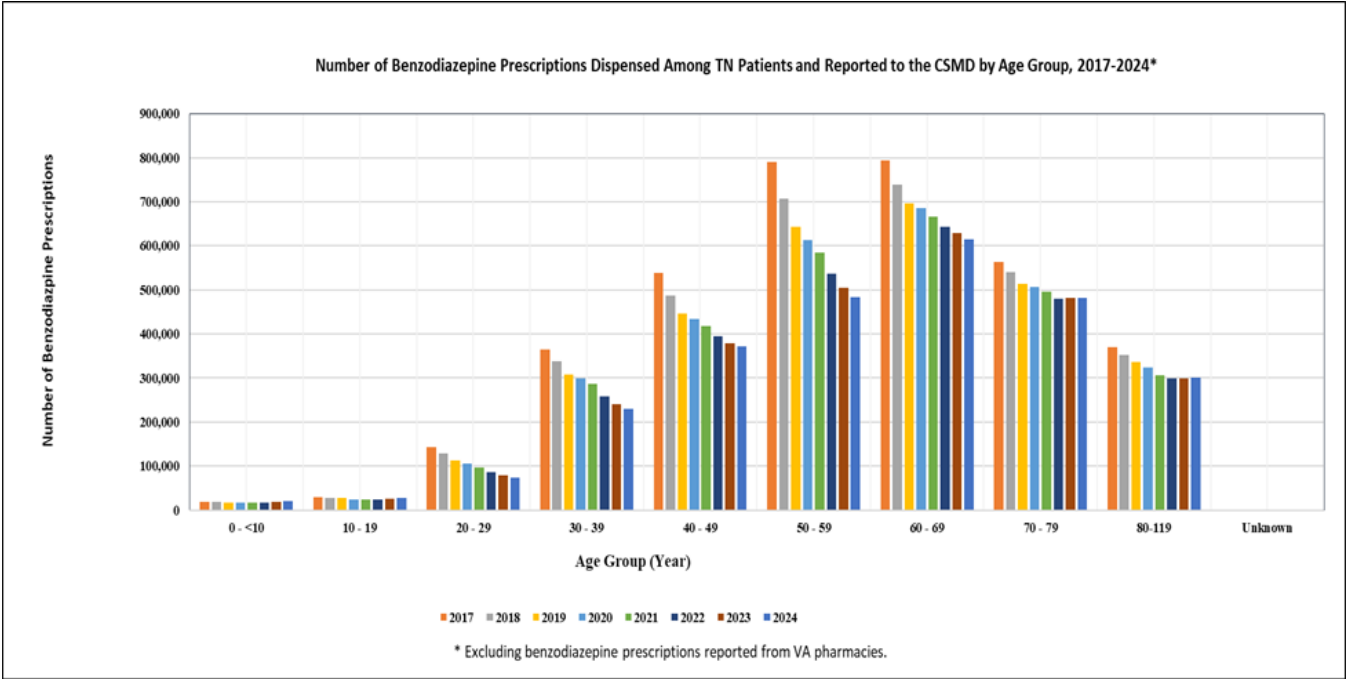
Change in MME Dispensed Among TN Patients by Age Group, 2017 vs. 2024



Trends Related to Utilization of Benzodiazepines

Benzodiazepines, such as alprazolam and diazepam, demonstrated a 28% decrease in prescriptions from 2017 to 2024. For 2024, this class has seen a decline in prescribing and dispensing for people between 20 and 69 years of age.

Number of Benzodiazepine Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Age Group, 2017-2024



Number of Benzodiazepine Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Age Group, 2017-2024*

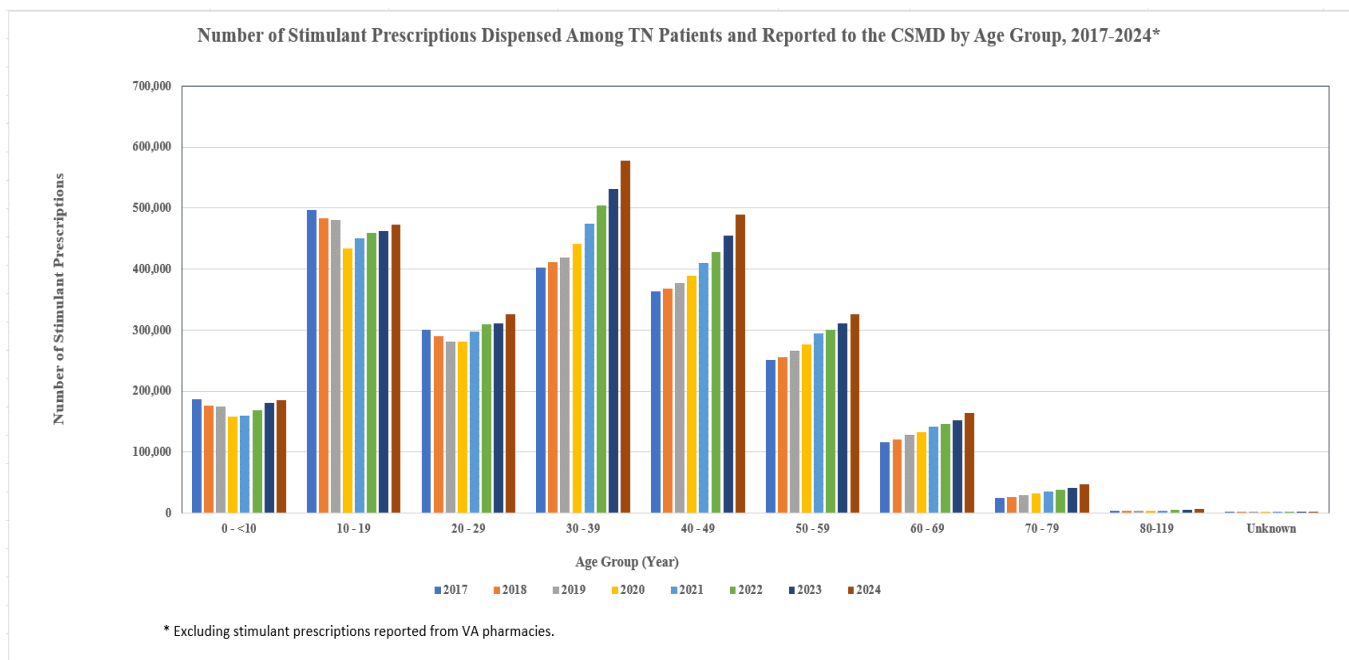
Age Group (Year)	2017	2018	2019	2020	2021	2022	2023	2024
0 - <10	18,107	18,186	17,765	16,614	17,028	17,309	18,847	19,956
10 - 19	28,656	27,457	27,049	24,992	24,499	24,407	25,637	26,957
20 - 29	143,631	128,716	113,586	105,501	97,054	85,650	79,226	74,594
30 - 39	365,518	338,388	307,374	299,596	286,261	258,253	240,628	229,391
40 - 49	538,131	487,801	446,427	433,812	417,953	395,476	378,285	370,944
50 - 59	791,029	707,126	642,404	612,978	583,798	537,292	505,644	482,966
60 - 69	793,941	739,114	695,843	685,679	666,540	643,456	629,263	614,875
70 - 79	562,873	539,494	514,237	506,404	495,932	480,763	481,037	481,541
80-119	370,694	352,891	335,525	324,388	306,546	299,294	299,334	300,783
Unknown	6	1	9	24	18	6	4	16

* Excluding benzodiazepine prescriptions reported from VA pharmacies.

Trends Related to Utilization of Stimulants

The number of prescriptions for stimulants has continued to increase, growing by almost 20.8% for patients in TN from 2017 to 2024.

Number of Stimulant Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Age Group, 2017-2024



Number of Stimulant Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Age Group, 2017-2024*

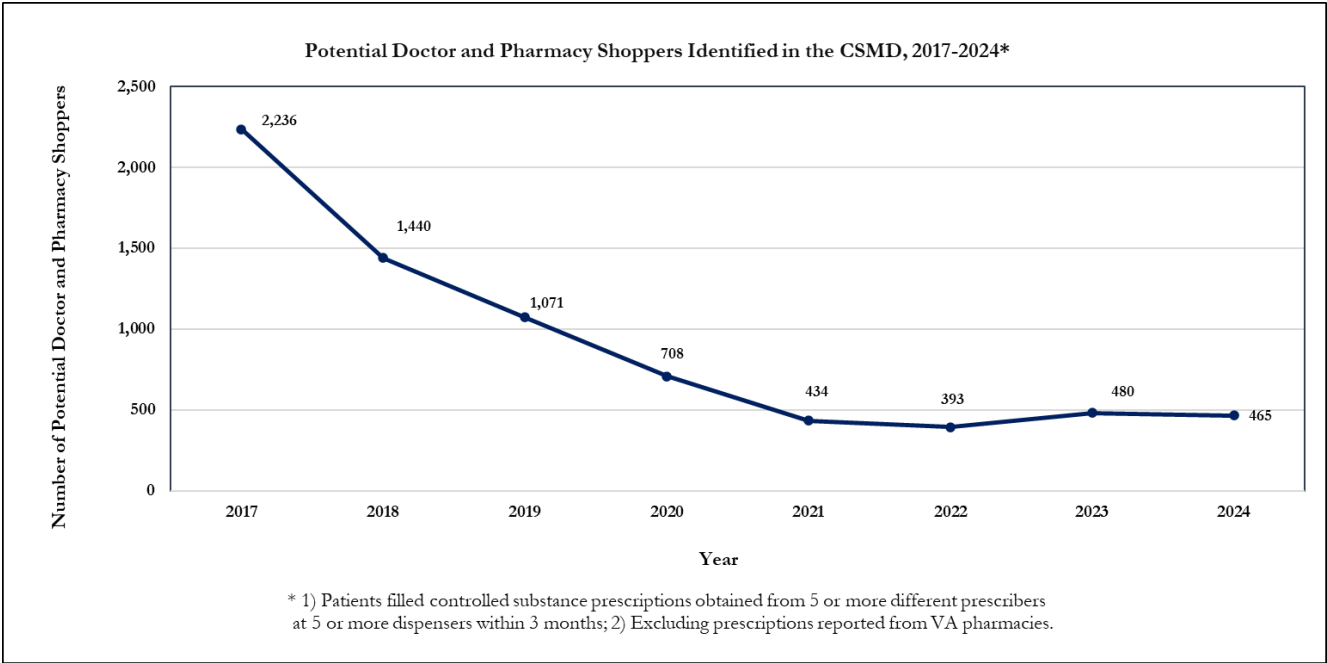
Age Group (Year)	2017	2018	2019	2020	2021	2022	2023	2024
0 - <10	186,502	176,274	174,814	157,681	160,033	168,184	179,872	184,804
10 - 19	496,235	484,055	480,135	434,407	449,977	459,185	462,493	472,635
20 - 29	300,694	290,716	281,486	281,337	298,114	309,468	311,138	325,478
30 - 39	401,920	410,948	419,563	441,051	474,810	504,532	531,002	577,069
40 - 49	363,674	368,637	377,194	388,316	410,199	427,674	454,549	489,888
50 - 59	251,716	255,635	265,316	276,351	293,989	300,513	311,163	325,281
60 - 69	116,857	120,832	127,641	133,359	141,124	146,111	152,462	163,406
70 - 79	24,192	26,706	29,487	31,499	34,653	37,667	41,883	46,612
80-119	3,899	4,187	4,423	4,263	4,378	4,652	5,472	6,318
Unknown	2	3	4	7	5	5	4	9

*** Excluding stimulant prescriptions reported from VA pharmacies.**

Decline in Potential Doctor and Pharmacy Shopping

TDH has defined a potential doctor and pharmacy shopper as an individual visiting five (5) or more prescribers and five (5) or more dispensers in a 90 day (3 months) period, referred to as “5-5-3 criteria”. Within TN, there has been an 79% decrease of potential doctor and pharmacy shopping patients from 2017 to 2024.

Potential Doctor and Pharmacy Shoppers Identified in the CSMD, 2017-2024



Gateway Electronic Health Record (EHR)/Pharmacy Management System Clinical Workflow Integration

The Gateway integration service allows healthcare providers the ability to view their patient’s history of controlled substance prescriptions, along with the Clinical Risk Indicators (CRI), within their clinical application (EHR/Pharmacy Management System). Throughout 2024, the CSMD has experienced a significant increase (44.3%) in the number of healthcare providers that are integrated with Gateway. Of the healthcare providers using Gateway, many independent practices experience efficiencies in their workflow to meet statutory regulations of the CSMD with no added financial burden.

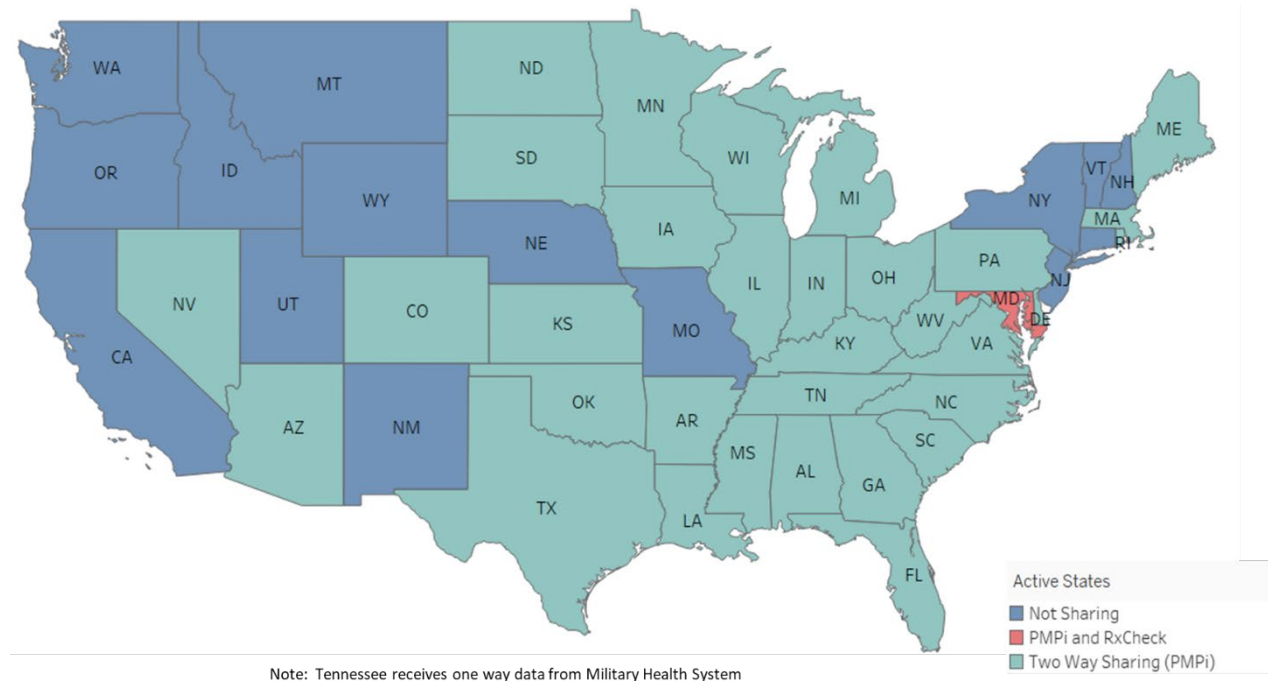
The CSMD team works diligently with the vendor to continue providing a stable environment for healthcare providers. The CSMD system uptime was 99.9% for 2024. One attribute of the CSMD system is the less than one-second response time when a patient request is initiated, given that the request does not include data from another state.

Increased Interstate Data Sharing

The CSMD program now shares data with many states in addition to the entire southeast United States. Each state or jurisdiction has unique regulations and requirements that require collaboration to share data to address regulatory compliance. The CSMD program utilizes, at no cost to the State, the National Association of Boards of Pharmacy Prescription Monitoring Program InterConnect (PMPi) to allow TN to share data. The PMPi is a highly secure communications exchange platform that facilitates the transmission of PDMP data across jurisdictions to authorized requestors, while ensuring that each PDMP’s data-access rules are enforced.

In 2021, TDH connected the CSMD with RxCheck, an interstate data sharing and integration system developed with support from the Bureau of Justice Assistance (BJA) and designed to complement current CSMD activities. It is of no cost to jurisdictions or CSMD users and allows jurisdictions to have an additional option for querying prescription data across jurisdictions. RxCheck also provides functionality for integration with EHRs using nationally recognized interoperability standards. RxCheck is governed jointly by the Integrated Justice Information Systems Institute (IJIS),

BJA, and a board of PDMP administrators in connected jurisdictions. Currently, BJA has made a commitment to fund RxCheck and support both the jurisdictions and governing body with training and technical support. TDH continues to work with other jurisdictions on interstate data sharing using RxCheck. Also, TDH continues to work on a pilot using RxCheck for EHR integration with CSMD data. The figure below provides a diagram of our interstate data partnerships with other jurisdictions for both PMPi and RxCheck:



Security Measures

In order to ensure that only those individuals and entities authorized pursuant to the PSA of 2016 have access to the information contained in the database, the CSMD employs the following security measures:

- All entities and individuals who are authorized to access the database in accordance with Tenn. Code Ann. §§ 53-10-306(a)(1)-(6), and (13) must first undergo a registration process during which the CSMD validates their identifying credentials and generates a unique username with instructions to create a password. As of January 26, 2022, all healthcare practitioners registered with the database must maintain a valid e-mail address associated with their database user profile. For healthcare practitioner delegates, an additional approval from their supervising healthcare practitioner is required.
- Any access by a quality improvement committee, as defined in Tenn. Code Ann. §§ 63-1-150 and 68-11-272, must strictly adhere to the requirements of Tenn. Code Ann. §§ 53-10-306(a)(9) and (10). Notably, the quality improvement committee cannot share any information obtained from the database.
- Before the Office of Inspector General, the Medicaid Fraud Control Unit, and the TennCare personnel identified in Tenn. Code Ann. § 53-10-306(a)(7) are given access to the database, they must submit a written request approved by their supervisor. The CSMD staff verifies each requester's credentials before issuing a unique, individual username with instructions to create a password. TDH and the CSMD Committee have partnered with TennCare to provide data sets which are subject to different

security protocols as required by TennCare's agreements with various vendors. The data sets provided to TennCare are defined data sets limited to TennCare recipients.

- The CSMD staff has oversight of the data accessed, updated, or viewed by all users through the creation of an audit trail for each user.
- Law enforcement personnel, as defined by Tenn. Code Ann. § 53-10-302, has two methods to obtain information sent to, contained in, and reported from the database in accordance with Tenn. Code Ann. § 53-10-306(a)(11). The first method is a paper process whereby law enforcement personnel submits a written request with a case number corresponding to a criminal investigation. Before releasing any information, CSMD staff confirms that the requester is on the approved list as required by Tenn. Code Ann. § 53-10-306(a)(11). The second method is via electronic registration and approval. Once electronically registered, law enforcement personnel can obtain the information directly from the CSMD web portal. Both methods create an audit trail.
- Requests for access by persons other than those individuals outlined in Tenn. Code Ann. §§ 53-10-306(a)(1)-(7), (9), and (10) are reviewed by Board of Pharmacy staff and TDH's Office of General Counsel (OGC) to determine whether their requests should be granted. The OGC also reviews all subpoenas and court orders to ensure compliance with the law before releasing any information from the database.
- In 2016, TDH expanded its internal access systems as part of the creation of the TDH IDS, which is designed to efficiently provide usable data access to a limited number of authorized users for various TDH data systems. The security and access related to this project is handled by various partners including Strategic Technology Solutions (STS), OGC, and OIA, in conjunction with oversight from the participating data source programs. The TDH IDS, which maintains certain CSMD data, resides in the State Data Center and is behind the State network firewall that prevents any access outside of the state firewall without the proper approved connection through the state's Virtual Private Network. All data on these servers are encrypted.
- Currently, only administrators and a select group of individuals who have been granted authority by the CSMD program have access to the CSMD data made available through the TDH IDS. Authorized users of the IDS have to receive permission from the Director of the OIA and the Director of the TN CSMD Program to access CSMD data from the TDH IDS.

TDH Grants Update

Over the last nine years, TDH has been awarded a number of federal grants aimed at building capacity for public health surveillance and prevention of drug overdose. Recognizing the important role that prescribers and PDMPs play in both surveillance and prevention activities, most of these grants have included activities intended to enhance the CSMD directly or to better utilize CSMD data in prevention, planning, and enforcement. The following grants all support enhancements to the CSMD or analytics projects using data from the CSMD:

- In September 2015, TDH was awarded a grant from the CDC to assist with funding epidemiologic studies pertaining to the nation's Prescription Drug Overdose (PDO) epidemic, enhance and maximize the CSMD, and develop community, insurer, and health system interventions. PDO: Prevention for States (PfS) funding was initially awarded to sixteen states and expanded upon the work already under way through the earlier "PDO: Boost" grant. The purpose of the PfS grant was to provide state health departments with additional resources and support needed to advance interventions for preventing prescription drug overdoses within

their own jurisdictions. The PfS grant ended in 2019 with many of the activities continuing under the Overdose Data to Action (OD2A) and Overdose Data to Action in States (OD2A-S) grants ([see below](#)).

- In 2016, TDH was awarded a grant from the Bureau of Justice Assistance (BJA) under the Harold “Hal” Rogers program to create rapid data-based collaboration between TDH, TBI, and the TDMHSAS. This grant, which ended in 2020, funded improved access for law enforcement and drug courts to the CSMD. It also supported the collection and integration of law enforcement and mental health data to better identify and react to emerging and existing hotspots, as well as changes in the overdose epidemic. Under the Hal Rogers funding, TDH began convening biweekly meetings of stakeholders from TDH, TBI, and TDMHSAS other state agencies to discuss ongoing activities and trends in the data.
- In 2017, TDH received another CDC grant to enhance surveillance of opioid overdoses, called Enhanced Surveillance of Opioid Overdose in States (ESOOS). For this grant, TDH worked to expand nonfatal overdose data gathering from TDH’s syndromic surveillance system; the Electronic Surveillance System for the Early Notification of Community Epidemics (ESSENCE). TDH also gathered and submitted expanded information on fatal overdoses in collaboration with the Office of the State Chief Medical Examiner. These fatal overdose data were submitted to a nationwide system called the State Unintentional Drug Overdose Reporting System (SUDORS). SUDORS data submitted under the ESOOS grant contained detailed information on toxicology and scene investigation for opioid overdose deaths in Tennessee SUDORS is a part of the National Violent Death Reporting System, also sponsored by the CDC.
- In 2018, TDH received two grants from the Department Of Justice’s BJA Comprehensive Opioid Abuse Program (COAP). One of these grants funded work to connect the CSMD to RxCheck and partially supported predictive modeling of overdose risk, conducted by VUMC. The other grant funds the integration of EMS overdose data into TDH’s IDS, expands overdose and drug surveillance to other substances of concern, and continues the interagency collaborations started under the 2016 Hal Rogers funding.
- In 2019, the CDC combined activities from previous opioid-focused grants into a single, more expansive grant opportunity called OD2A. In addition to the surveillance activities undertaken for the ESOOS grant and the CSMD-related data work for PfS, the OD2A grant expanded the funding available for overdose prevention activities across the state. The TDH Overdose Response Coordination Office has been consistently working closely with regional and metro health departments across the state to start or expand local prevention activities, academic detailing and opioid and overdose education for prescribers, and treatment resource locators.
- In 2020, TDH was awarded for three years under the FY2020 Hal Rogers grant opportunity to continue enhancements to the CSMD and related data systems. These funds are used to extend the amount of time that TDH can offer support for Gateway workflow integration to CSMD users and expand the use of RxCheck for EHR integration. Funds also used to improve the system for receiving drug shipment data from distributors and wholesalers. Another activity supported by this grant continues and expands integration of data from the TBI with TDH data.
- In 2022, TDH was awarded for three years under the FY2022 Hal Rogers grant opportunity to continue enhancements to the CSMD and related data systems. These funds will be used to continue RxCheck interstate data sharing onboarding and pilot an additional facility for EHR integration. Funds will provide support for the hosting and maintenance of the enhanced application for submission of controlled drug shipment data. Funds will also be used to develop an in-depth Education program for CSMD system users. Additionally, the funds will be used for the enhancement of public health and public safety to build a framework for information sharing to address the drug overdose epidemic is another activity supported by this grant.
- In 2023, TDH was awarded five years of funding for the OD2A-S grant. This grant expands work from the previous OD2A grant with additional capacity for a strategy focusing on data linkage. In this strategy, CSMD

data will be linked to fatal and nonfatal overdose data to assess trends over time. OIA and ORCO work closely to ensure grant strategies are completed.

- In 2024, the Tennessee Department of Health (TDH) secured a two-year grant through the FY2024 Hal Rogers program to advance enhancements to the Controlled Substance Monitoring Database (CSMD) and related data systems. This funding will support key initiatives such as expanding RxCheck interstate data sharing, piloting an additional Electronic Health Records (EHR) integration facility, and maintaining the enhanced system for controlled drug shipment data submission. Additionally, the grant will facilitate the development of a comprehensive education program for CSMD users and strengthen public health and safety partnerships to improve information sharing in the fight against the drug overdose epidemic.

TDH Recommends the Following Approaches to the Prescription Drug Use and Misuse

1. Prevent opioid and other drug use and misuse through an effective and comprehensive approach to prevention.

- Implement effective public education/awareness campaigns.
- Ensure that schools and communities implement effective prevention initiatives.
- Reduce availability of and accessibility to addictive opioids and other controlled substances.
- Utilize Prescription Drug Monitoring Programs (PDMPs).
- Participate in Safe Prescribing Initiatives for pain management.
- Utilize the Tennessee Chronic Pain Guidelines (2024 version) located on the TDH website at: <https://www.tn.gov/content/dam/tn/health/healthprofboards/pain-management-clinic/ChronicPainGuidelines.pdf>.
- Support and maximize use of Prescription Take-Back Programs and related options.
- Continue to regulate Pain Management Clinics. (See most recent report on Prescription Drug Abuse and Pain Management Clinics at <https://www.tn.gov/health/health-program-areas/statistics/tn-doh-publications.html>)

2. Reduce overdose deaths and other harmful consequences through harm reduction strategies.

- Increase community access to naloxone and fentanyl testing strips through prevention partners at TDMHSAS and TDH health departments.
- Increase the number of Syringe Services Programs (SSP).
- Monitor and react rapidly to emerging drug trends, prescription or illicit.

3. Improve opioid addiction treatment through an effective and comprehensive approach.

- Increase treatment capacity.
- Increase availability of MAT and Learn how to administer/prescribe MAT. (<https://www.samhsa.gov/medications-substance-use-disorders>).
- Provide comprehensive recovery support services following treatment.
- Work to alleviate the stigma associated with seeking treatment for substance use disorder.

In summary, the drug epidemic continues to have a major impact on the State of Tennessee, taking more lives than motor vehicle accidents and suicides combined. Real, effective solutions will require a shift in the way TDH thinks

about and respond to substance use disorders, including our abilities to prevent, treat, destigmatize, and advance harm reduction through the dissemination of evidence-based practices.

Conclusion

The CSMD, along with the Gateway integration, is a valuable service to inform healthcare providers of a patient's controlled substance prescription history before, during, and after a treatment plan has been decided. The CRI displayed on the reports inform healthcare providers of potential high risk patients. The information in the CSMD proves to be useful in improving patient safety and quality of care.

The epidemic of opioid use and misuse has continued to change. Overdose deaths continue to increase; some involving illicit opioids, mostly fentanyl. TDH is working hard to better understand data from various reporting systems and work collaboratively with other agencies to improve prevention opportunities. Data from the CSMD are at the center of numerous innovative efforts using TDH data to understand and predict overdose risk, educate patients and prescribers, decrease inappropriate prescribing, and improve access to treatment.

A multi-faceted approach, including prevention of exposure to opioids and other controlled substances, early diagnosis and treatment of Substance Use Disorder (SUD), and the life-saving work of law enforcement to reduce availability of illicit, addictive substances are all needed to reverse the trend. While actions at the community level are most impactful, intervention is also needed at the state and federal levels. With the removal of the Drug Addiction Treatment Act Waiver (DATA-Waiver or X-Waiver), treatment and prescriptions for SUD are increasing.

TDH would like to provide a special thanks to the current and past members of the legislature, the CSMD Committee, and the leadership of other federal and state agencies as we continue to work together in preventing harm to the public stemming from the drug epidemic.

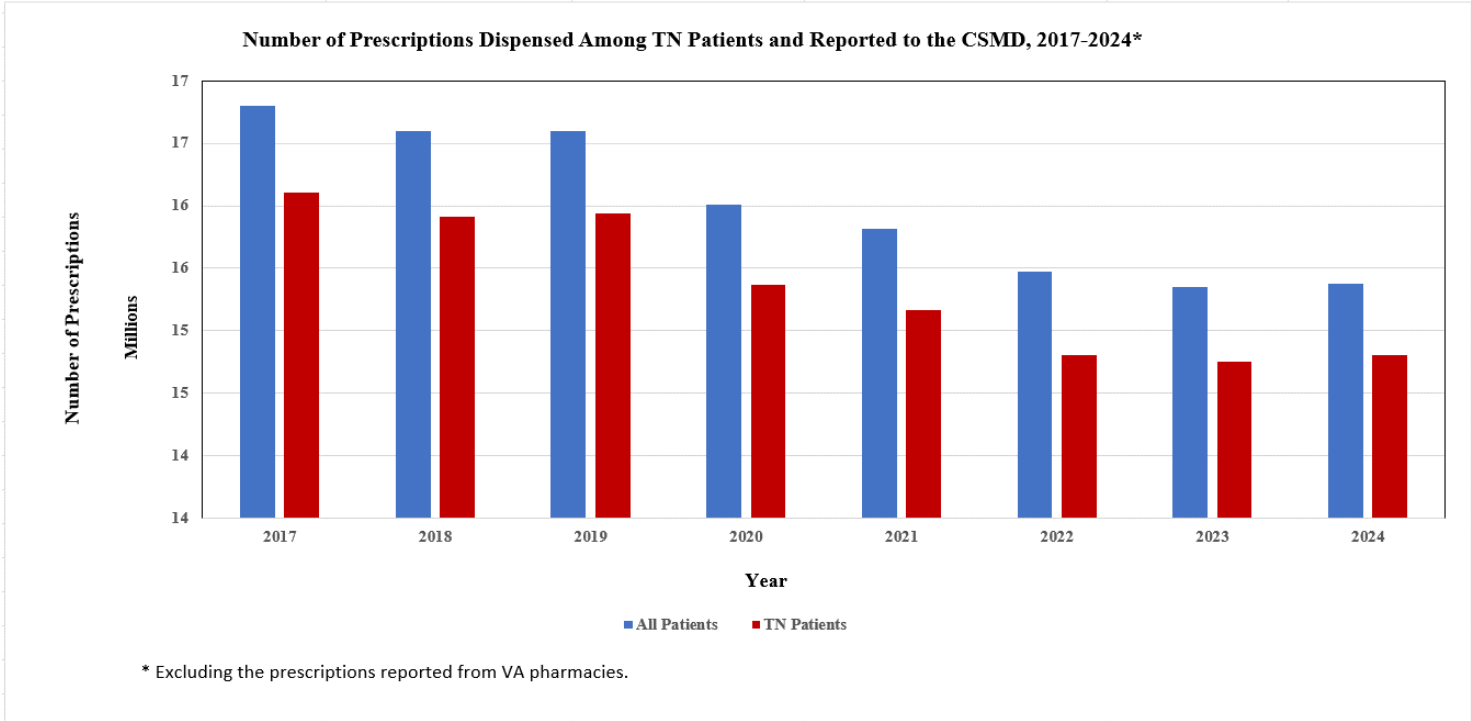
2024 Members of the CSMD Committee

Member Name	Board
Deborah Christiansen, M.D.	Board of Medical Examiners
Joseph Britton, D.D.S.	Board of Dentistry
Sheryl Wright, APRN	Board of Nursing
Zachary McCarthy, O.D.	Board of Optometry
Shant Garabedian, D.O.	Board of Osteopathy
Marlin Blane, Pharm.D., J.D.	Board of Pharmacy
Bhekumuzi Khumalo, D.P.M.	Board of Podiatry
Sarah Spidel, D.V.M.	Board of Veterinary Medical Examiners
Robert White, PA-C	Board of Physician Assistants
James Diaz-Barriga	Public Member Board of Medical Examiners
Jake Bynum	Public Member Board of Pharmacy

Appendix

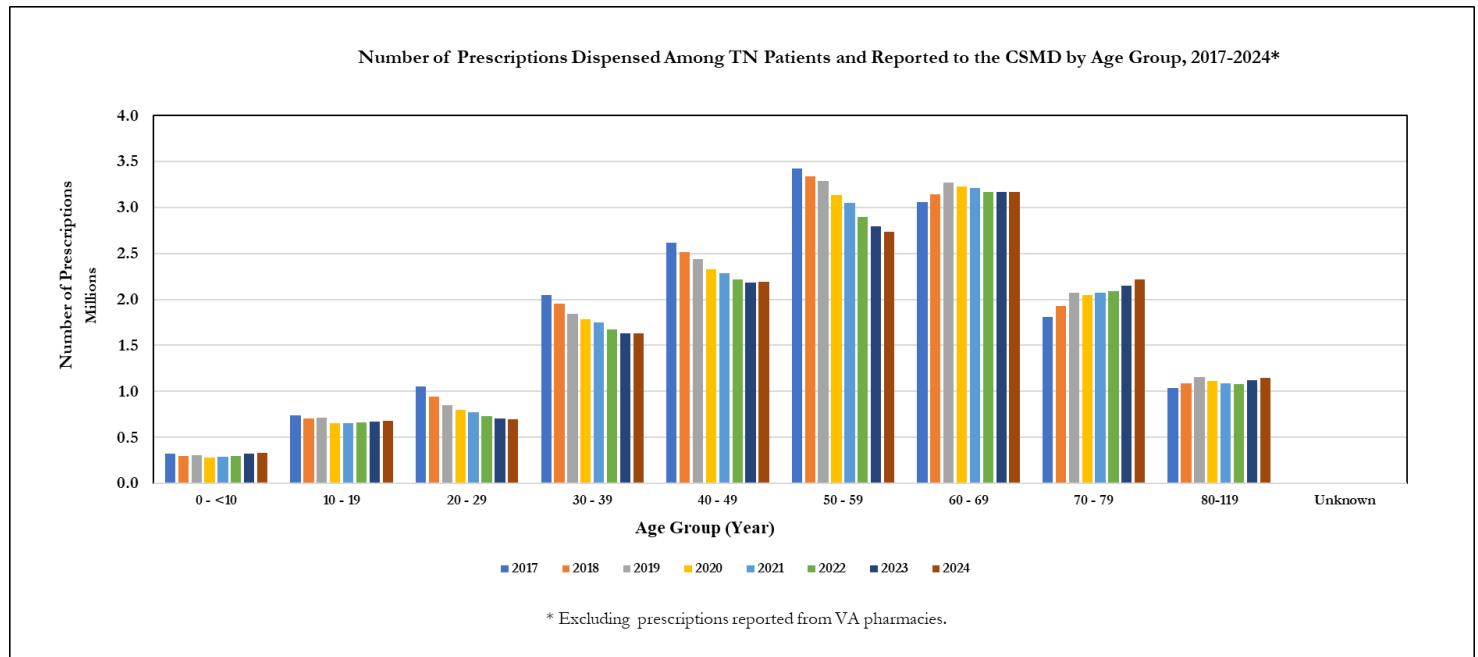
The CSMD data used for this report were downloaded on January 6, 2025. MME calculations were only limited to the opioid products that were listed in CDC's MME conversion tables from 2011 to 2021. The CDC has adjusted certain drug conversion factors over time for various reasons. If a drug had different MME conversion factors in different version tables, the data analysis provided through 2021 used the conversion factor provided in the latest version of the CDC table. Therefore, different MME results for a similar indicator would be expected for CSMD annual reports published in previous years. Prescriptions and MME identified for TN patients were based on a patient's state listed as 'TN' or state Federal Information Processing Standard (FIPS) code of '47' on the patient's address associated with a prescription. Otherwise, the patient was identified as a non-TN patient. Classification of controlled substances was based on combination of CDC tables and 2024 IBM Micromedex Red Book Select file. If a drug in the CSMD was not classified, the drug was classified as 'other' in this report. Due to improvements in the classification of prescriptions reported to the CSMD, some totals in this year's report may appear larger than in previous years. Stimulant counts have increased with the addition of Schedule III and IV stimulants, and gabapentin prescriptions are now classified separately. These prescriptions were previously classified in the 'other' category. Please note that human and animal prescription data are included in this report as it relates to the data analysis through 2024.

Number of Prescriptions Dispensed Among TN Patients and Reported to the CSMD, 2017-2024



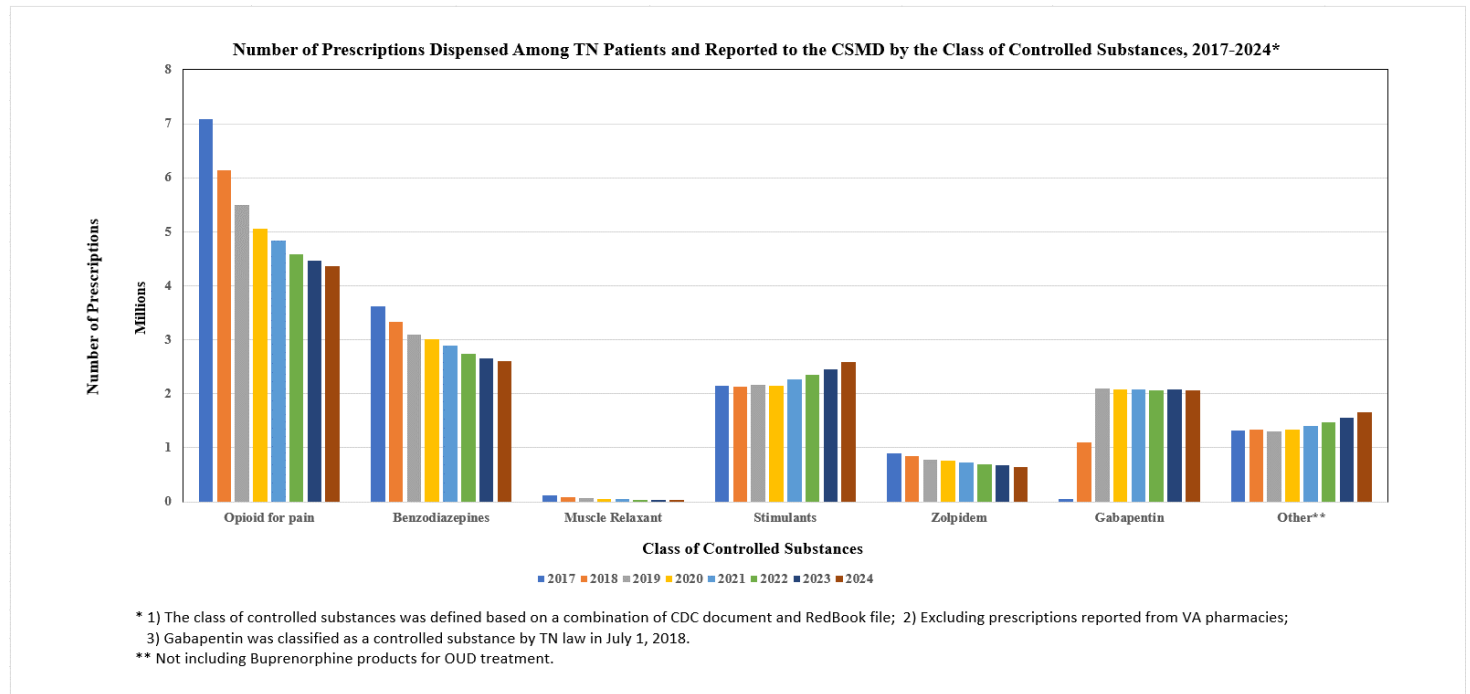
Number of Prescriptions Dispensed Among TN Patients and Reported to the CSMD, 2017-2024*				
Year	All Patients	Change (%)	TN Patients	Change (%)
2017	16,798,809	-	16,104,411	-
2018	16,596,898	-1.2	15,912,436	-1.2
2019	16,600,469	0.0	15,937,278	0.2
2020	16,009,471	-3.6	15,369,484	-3.6
2021	15,820,334	-1.2	15,169,607	-1.3
2022	15,472,577	-2.2	14,807,604	-2.4
2023	15,354,490	-0.8	14,751,654	-0.4
2024	15,378,999	0.2	14,802,257	0.3
*Excluding prescriptions reported from VA pharmacies.				

Number of Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Age Group, 2017-2024



Number of Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Age Group, 2017-2024*								
Age Group	2017	2018	2019	2020	2021	2022	2023	2024
0 - <10	319,059	299,881	303,237	280,759	285,649	298,631	322,586	334,275
10 - 19	735,237	708,459	715,532	650,405	657,515	664,456	672,787	682,222
20 - 29	1,053,286	942,459	850,859	795,081	774,540	732,134	707,735	697,820
30 - 39	2,050,944	1,953,875	1,845,572	1,780,191	1,752,093	1,670,293	1,630,769	1,630,502
40 - 49	2,612,163	2,513,738	2,435,430	2,331,098	2,283,327	2,217,368	2,186,594	2,194,838
50 - 59	3,426,029	3,336,631	3,283,295	3,138,329	3,048,966	2,892,932	2,796,420	2,733,326
60 - 69	3,060,342	3,141,129	3,273,248	3,229,694	3,210,644	3,165,511	3,165,353	3,165,603
70 - 79	1,811,815	1,926,445	2,074,099	2,050,410	2,072,914	2,086,364	2,151,768	2,219,757
80-119	1,035,512	1,089,786	1,155,767	1,113,149	1,083,663	1,079,659	1,117,371	1,143,450
Unknown	24	33	239	368	296	256	271	464
*Excluding prescriptions reported from VA pharmacies.								

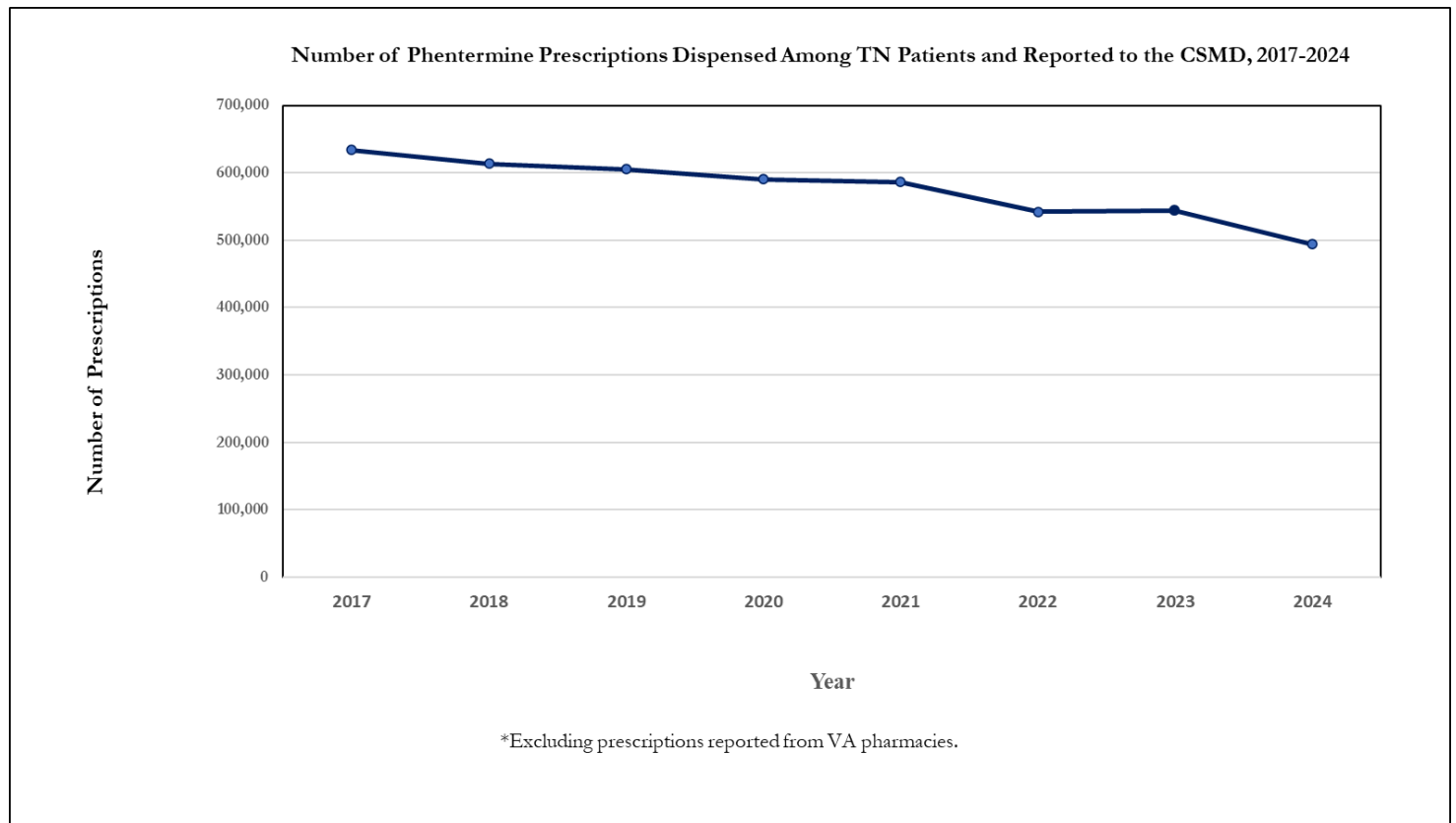
Number of Prescriptions Dispensed Among TN Patients and Reported to the CSMD by Class of Controlled Substances, 2017 – 2024



Year	Opioid for pain	Benzodiazepines	Muscle Relaxant	Stimulants	Zolpidem	Gabapentin	Other**
2017	7,076,353	3,612,586	117,493	2,145,691	901,040	60,720	1,321,263
2018	6,140,466	3,339,174	88,400	2,137,993	840,083	1,103,165	1,334,720
2019	5,493,193	3,100,219	65,327	2,160,063	786,711	2,103,903	1,306,271
2020	5,051,433	3,009,988	55,476	2,148,271	762,784	2,089,319	1,342,345
2021	4,838,040	2,895,629	48,743	2,267,282	730,003	2,086,344	1,400,560
2022	4,577,036	2,741,906	42,101	2,357,991	690,843	2,064,233	1,477,783
2023	4,461,151	2,657,905	36,648	2,450,038	670,584	2,072,991	1,555,744
2024	4,359,008	2,602,023	32,758	2,591,500	644,203	2,070,573	1,658,182

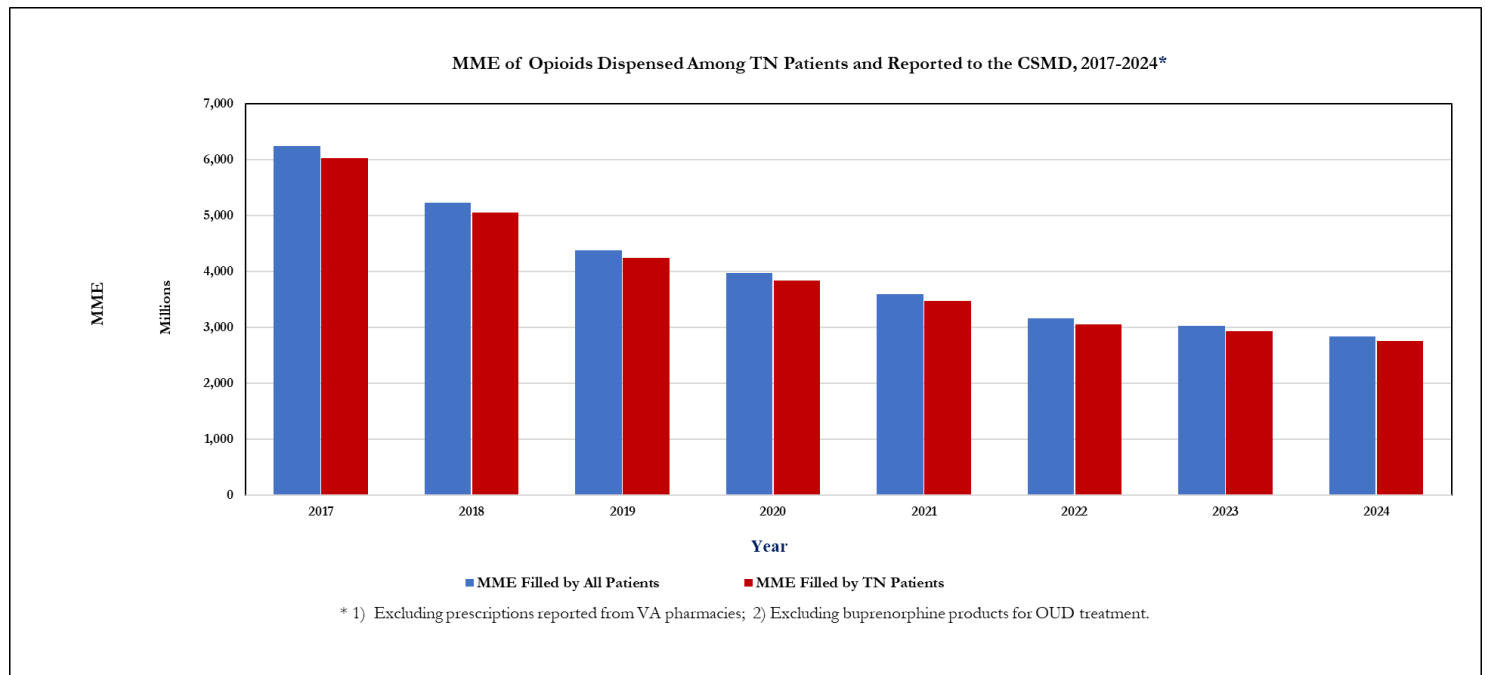
* 1) The class of controlled substances was defined based on a combination of CDC document and RedBook file; 2) Excluding prescriptions reported from VA pharmacies; 3) Gabapentin was classified as a controlled substance by TN law in July 1, 2018.

Number of Phentermine Prescriptions Dispensed Among TN Patients and Reported to the CSMD, 2017 – 2024



Number of Phentermine Prescriptions Dispensed and Reported to the CSMD, 2017 to 2024*		
Year	Total Number of Phentermine Prescriptions Reported to the CSMD	Change (%)
2017	633,873	-
2018	613,535	-3.2
2019	605,129	-1.4
2020	590,313	-2.4
2021	586,450	-0.7
2022	542,104	-7.6
2023	544,444	0.4
2024	493,936	-9.3
* Prescriptions from VA pharmacies were not included.		

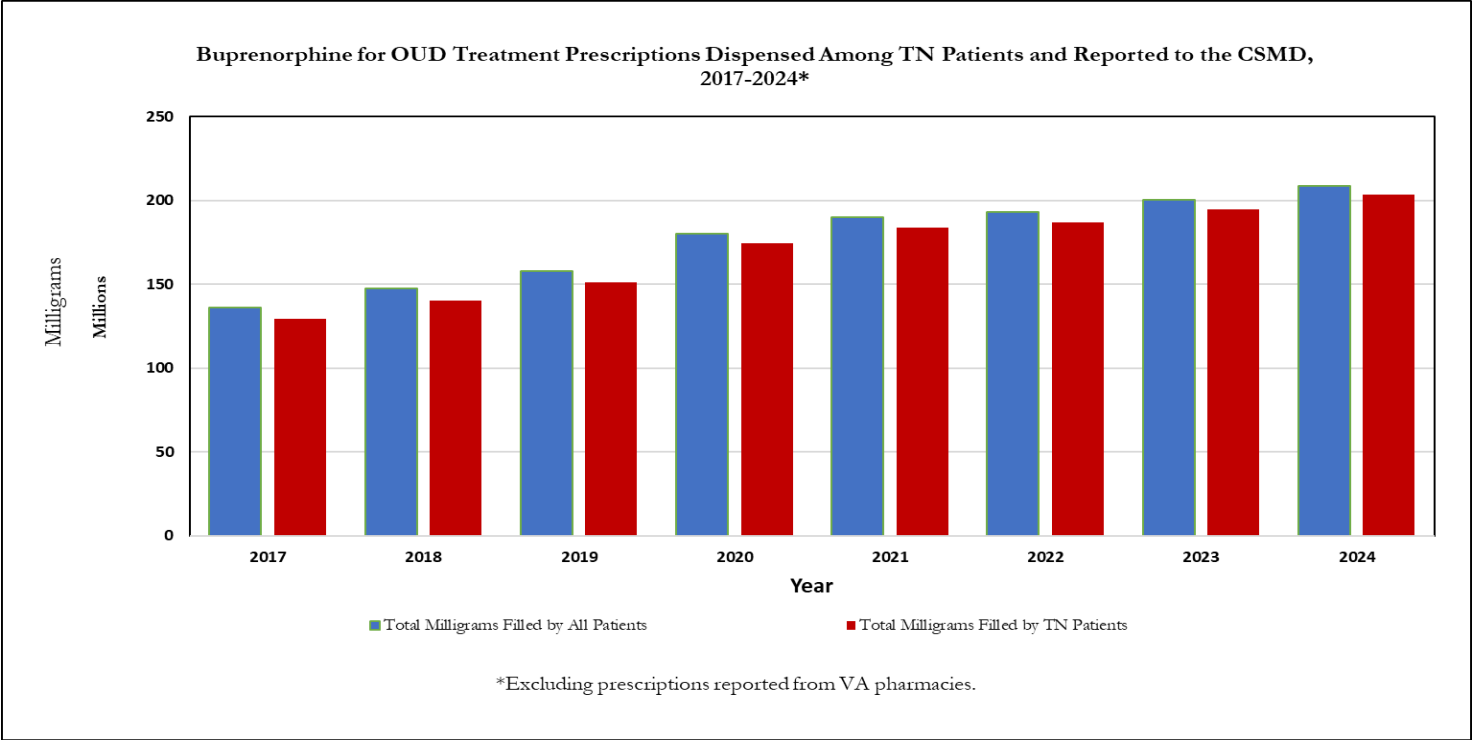
MME of Opioids Dispensed Among TN Patients and Reported to the CSMD, 2017-2024



MME of Opioids Dispensed Among TN Patients and Reported to the CSMD, 2017-2024*				
Year	MME Filled by All Patients	Change (%)	MME Filled by TN Patients	Change (%)
2017	6,239,324,036	-	6,019,445,631	-
2018	5,235,475,023	-16.1	5,057,124,576	-16.0
2019	4,383,132,970	-16.3	4,243,146,604	-16.1
2020	3,973,037,356	-9.4	3,839,878,755	-9.5
2021	3,589,263,433	-9.7	3,467,015,734	-9.7
2022	3,158,739,932	-12.0	3,052,995,345	-11.9
2023	3,027,390,156	-4.2	2,928,731,796	-4.1
2024	2,843,381,880	-6.1	2,756,113,503	-5.9

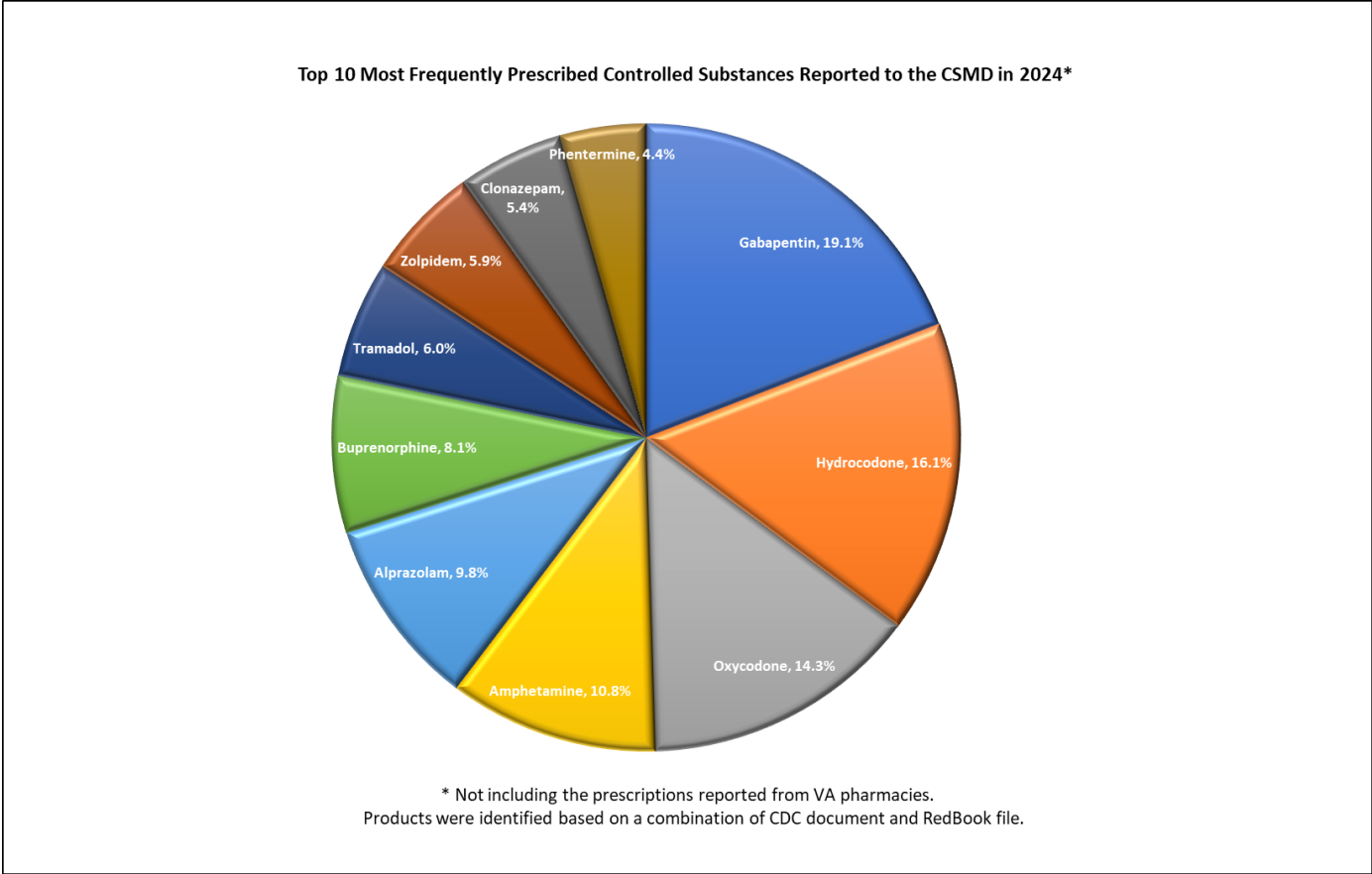
* 1) Excluding prescriptions reported from VA pharmacies; 2) excluding buprenorphine products for OUD treatment.

Buprenorphine for OUD Treatment Prescriptions Dispensed Among TN Patients and Reported to the CSMD, 2017-2024



Buprenorphine for OUD Treatment Prescriptions Dispensed to Among TN Patients and Reported to the CSMD by Age Group, 2017-2024*								
Age Group (Year)	2017	2018	2019	2020	2021	2022	2023	2024
0 - <10	7,573	4,976	6,912	4,592	3,583	1,248	688	112
10 - 19	150,728	85,961	76,629	93,752	77,514	85,550	79,954	83,487
20 - 29	24,495,561	22,830,789	20,574,068	19,852,157	16,861,877	13,855,819	11,924,362	9,734,647
30 - 39	51,698,816	54,914,525	58,015,633	66,566,036	68,177,571	65,985,953	65,041,450	64,246,361
40 - 49	30,274,624	34,783,619	39,294,419	47,064,289	52,562,044	56,757,259	62,268,474	68,098,072
50 - 59	16,755,930	20,070,475	23,325,941	27,483,918	30,175,819	31,867,841	33,969,530	36,639,180
60 - 69	5,149,010	6,900,436	8,829,124	11,651,690	14,130,589	16,178,677	18,474,131	20,758,495
70 - 79	595,587	816,672	1,140,072	1,564,710	1,871,600	2,212,214	2,819,249	3,444,540
80-119	38,018	51,838	54,778	96,275	121,591	146,084	195,180	232,263
Unknown	0	0	0	336	0	224	144	540
* Excluding prescriptions reported from VA pharmacies.								

Top 10 Most Frequently Prescribed Controlled Substances Reported to the CSMD in 2024



Number of Top 8 Prescriptions in 'Other' Group Dispensed to TN Patients and Reported to the CSMD, 2017-2024*

Year	Atropine/Difenoxin	Esterified Estrogen / Methyltestosterone	Eszopiclone	Lacosamide	Phenobarbital	Pregabalin	Testosterone	Zaleplon
2017	60,941	22,407	91,756	33,721	72,681	346,725	227,117	22,237
2018	57,376	20,219	95,120	35,836	70,894	349,897	234,775	20,860
2019	55,164	18,057	98,571	38,350	69,896	355,783	251,174	20,015
2020	50,243	16,541	105,470	41,131	67,598	389,082	283,469	19,855
2021	51,455	14,982	109,966	43,869	66,287	411,439	312,623	19,824
2022	51,736	13,581	113,693	46,705	59,611	428,275	352,246	18,834
2023	45,236	12,379	119,157	52,619	60,028	453,121	343,265	19,329
2024	43,559	11,649	125,824	61,494	58,949	479,543	393,628	19,236

* 1) The class of controlled substances was defined based on RedBook file; 2) Excluding prescriptions reported from VA pharmacies.

Acronyms

Bureau of Justice Assistance	BJA
Centers for Disease Control and Prevention	CDC
Clinical Risk Indicators	CRI
Comprehensive Opioid Abuse Program	COAP
Controlled Substance Monitoring Database	CSMD
Department of Justice	DOJ
Drug Addiction Treatment Act	DATA
Drug Overdose Reporting	DOR
Electronic Health Record	EHR
Electronic Surveillance System for the Early Notification of Community Epidemics	ESSENCE
Emergency Medical Services	EMS
Enhanced Surveillance of Opioid Overdose	ESOOS
Federal Information Processing Standard	FIPS
Integrated Data System	IDS
Integrated Justice Information Systems Institute	IJIS
Medication-Assisted Treatment	MAT
Medication for Opioid Use Disorder	MOUD
Morphine Milligram Equivalents	MME
Neonatal Abstinence Syndrome	NAS

Office of General Counsel	OGC
Office of Informatics and Analytics	OIA
Opioid Use Disorder	OUD
Overdose Data to Action	OD2A
Overdose Data to Action States	OD2A-S
Prescription Drug Monitoring Program	PDMP
Prescription Drug Overdose	PDO
Prescription Monitoring Program InterConnect	PMPi
Prescription Safety Act	PSA
Prevention for States	PfS
State Unintentional Drug Overdose Reporting System	SUDORS
Strategic Technology Solutions	STS
Substance Use Disorder	SUD
Syringe Services Programs	SSP
Tennessee	TN
Tennessee Bureau of Investigations	TBI
Tennessee Department of Health	TDH
Tennessee Department of Mental Health and Substance Abuse Services	TDMHSAS
Vanderbilt University Medical Center	VUMC
Veterans Administration	VA
Veterans Health Administration	VHA



Tennessee Controlled Substance Monitoring Database
Director, Tennessee Controlled Substance Monitoring Database
665 Mainstream Drive, 2nd Floor Nashville, Tennessee 37243
<https://www.tn.gov/health/csmd/>